

Mina Malak Habib

Backend Developer | Software Engineer

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Profile

Computer Engineering graduate from Helwan University with strong skills in **backend development** using **Java**, **Python**, and **C++**. Proficient in **Spring Boot**, **RESTful APIs**, and database management with **MySQL** and **PostgreSQL**. Passionate about building **scalable**, **secure**, and high performance backend systems while writing **clean**, **maintainable code** and **collaborating** effectively in agile teams.

Education

Bachelor of Computer and Systems Engineering, *Helwan University* 09/2020 – 07/2025
• **Overall Grade:** Very Good with High Honors (Ranked 26th in class) Cairo Egypt
• **Graduation Project:** Excellent

Work Experience

Open-Source Applications Developer (Full-Stack Trainee), 10/2025 – Present
National Telecommunication Institute (NTI) Smart Village, Egypt
Developing full-stack web applications using Node.js, Express, React, Next.js & Vue.js with database design (MySQL, MongoDB) and deployment via Docker & GitHub Actions.

Languages

English	Arabic
Proficient	Native

Courses

Introduction to Databases for Back-End Development, *Meta Back-End Developer Specialization, Coursera*
• Completed course covering SQL, database design, and data management

Version Control, *Meta Back-End Developer Specialization, Coursera*
• Completed a hands-on course covering Git, GitHub, and collaborative version control workflows.

Programming in Python, *Meta Back-End Developer Specialization, Coursera*
• Covering Python syntax, control flow, data structures, OOP, error handling, and unit testing through hands-on coding and problem-solving projects.

Introduction to Back-End Development, *Meta Back-End Developer Specialization, Coursera*
• Gained foundational skills in HTML, CSS, Bootstrap, React, and built a responsive webpage project.

Database Design Fundamentals, *Mahara tech*
• ER diagrams, normalization, relational design, SQL queries.

Object-Oriented Programming in C++, *Mahara tech*
• OOP principles, inheritance, polymorphism, operator overloading, templates.

Internet Of Things (IOT) Application Development, *Information Technology Institute*
• IoT value chain, methodologies, hands-on labs with IoT devices.

Basic Embedded Systems, *Information Technology Institute*
• Covered embedded concepts, C/Embedded C, data structures, AVR drivers, RTOS, hardware labs.

Projects

University Management System Project, *Using Java*

Developed a Java-based system with CRUD operations, data management for students, professors, and admins.

A Fully Automated Car (AVR, USART, LCD, Sensors)

Designed a smart car with ultrasonic sensor, LCD display, and dual microcontroller communication for control and monitoring.

Smart Home System Using IOT (ESP32, Raspberry Pi, MQTT)

Built a multi-node IoT setup for temperature/humidity monitoring using ESP-NOW and MQTT integration for real-time data visualization.

ADC Implementation for ARM7 Processor Using Assembly

Implemented ADC driver in ARM7 assembly with interrupt-driven data acquisition for real-time signal processing.

Personalized Product Recommendation System using Amazon product reviews

Created a sentiment-based recommendation model using Amazon reviews dataset with Naive Bayes and text preprocessing.

Skills

Technical Skills

- **Programming Languages:** Python, Java, JavaScript (Node.js) or similar
- **Frameworks:** Django, Spring Boot, Express.js, or Nest.js
- **Databases:** MySQL, MongoDB • **API Development:** RESTful & GraphQL APIs
- **Version Control:** Git, GitHub, CI/CD pipelines (Jenkins, GitHub Actions)

Soft Skills

- Communication and Presentation
- Team Collaboration and Leadership
- Problem-Solving and Critical Thinking
- Time Management and Organization

Graduation Project

Automotive Protective Guard (APG): Developed a real-time IDS for vehicles using Raspberry Pi 5 with CAN, Wi-Fi, and Bluetooth monitoring modules (MCP2515, Ubertooth, TP-Link).